

Energy-Guided Denoising Contrastive Learning for Molecular Property Prediction

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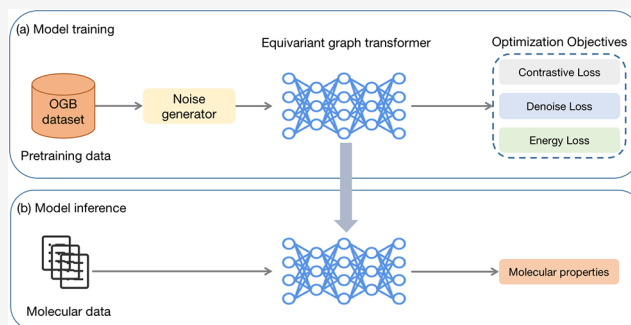
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ABSTRACT: Self-supervised coordinate denoising presents a promising approach to learning 3D molecular representations for predicting molecular properties. However, existing methods often rely on random perturbations, which overlook the heterogeneous chemical environments and may lead to physically unreasonable structures. To address this limitation, we propose an energy-guided denoising contrastive learning framework with an adaptive noise generator. This generator produces structure-aware, atom-specific perturbations conditioned on local chemical environments, guiding the encoder to learn a meaningful representation of the molecular potential energy surface (PES). Our pretraining framework integrates three complementary components: (i) equivariant denoising, (ii) contrastive alignment between original and perturbed conformations, and (iii) self-supervised energy-gap prediction. Extensive evaluations demonstrate that our method achieves state-of-the-art performance across several molecular property prediction benchmarks. The case studies show that our method is interpretable and has learned the relationship between chemical and energy information. The source code is available at <https://github.com/wangjx22/EDCL>.



INTRODUCTION

Accurately predicting molecular properties from three-dimensional (3D) structures is a central challenge in computational chemistry and drug discovery, with far-reaching implications for molecule design and therapeutic development.¹ Recent advances in deep learning—especially Equivariant Graph Neural Networks (EGNNs) that exploit geometric symmetries—have achieved outstanding performance on 3D molecular tasks.² However, this progress is constrained by the scarcity and cost of high-quality labeled data. In contrast to computer vision (CV) and natural language processing (NLP), where self-supervised learning (SSL) on large unlabeled corpora is now routine for learning generalizable features,^{3,4} pretraining paradigms tailored to 3D molecular science remain comparatively underexplored.

Among the emerging pretraining approaches, coordinate denoising has gained attraction as a physically motivated route to 3D representation learning.⁵ The idea is to perturb atomic coordinates (e.g., masking, substitution and spatial shifts), and then train a model to reconstruct the original conformations from the perturbed ones. This procedure encourages the model to learn the molecular force field (the gradient of the potential energy) and is theoretically connected to denoising score matching.⁶ When well designed, such a task can guide the encoder to substantiate the physical regularities governing conformational stability and generate representations that reflect the Potential Energy Surface (PES).⁷

Despite these advantages, mainstream 3D denoising methods commonly rely on random or handcrafted perturbations such as uniform Gaussian noise or rule-based shifts.^{8,9} They ignore the pronounced heterogeneity of chemical environments within a molecule. For instance, a methyl side chain attached to a rigid aromatic ring can rotate relatively freely with minor energetic penalty, whereas an equivalent shift on an atom within the ring can disrupt conjugation, sharply increasing energy and yielding an implausible conformation. This one-size-fits-all strategy thus produces physically inconsistent training signals, diverting the model toward “correcting” artifacts rather than learning the true geometric-energetic landscape, and ultimately limiting the quality of learned representations.

To address this problem, in this paper we propose an energy-guided denoising contrastive learning method with an adaptive noise generator for learning molecule representations, which are then used for accurate molecular property prediction. Instead of injecting random noise, we employ a noise generator to produce atom-specific perturbations for each atom in a molecule. We

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embed this generator into a unified multitask objective designed to yield physically faithful and conformation-robust representations from complementary perspectives: (i) an equivariant denoising task that predicts corrective shifts to recover ground truth coordinates for each atom; (ii) a contrastive objective that aligns global representations of the same molecule before and after perturbation, promoting invariance to conformational changes; and (iii) energy-gap supervision that injects explicit physical priors. To prevent “generator collapse” (predicting near-zero noise), we regularize the predicted noise distribution toward a principled prior via a Kullback–Leibler (KL) divergence term, stabilizing training and preserving perturbation diversity.

In summary, our contributions are as follows:

- We propose a learnable generator that can generate adaptive noise to replace random noise, enabling a physically meaningful, atom-level self-supervision pre-training process.
- We develop a synergistic multitask objective pretraining paradigm that integrates equivariant denoising, contrastive alignment, and energy-gap supervision to learn robust and informative 3D molecular representations for molecular property prediction.
- We conduct extensive experiments on several molecular property prediction benchmarks including MoleculeNet and QM9, demonstrating the state-of-the-art performance of the proposed method, and further validate the effectiveness of its components by ablation and case studies.

RELATED WORK

With the increasing recognition of the importance of 3D geometry, self-supervised learning (SSL) for molecules has evolved along two main lines: contrastive learning (CL) and generative learning, where 3D coordinate denoising emerges as a typical generative strategy.

Contrastive Self-Supervised Learning

Contrastive learning (CL) is a class of self-supervision learning methods, usually used to learn discriminative representations by distinguishing positive pairs (e.g., two views of the same instance) from negative pairs (e.g., views of different instances). Typically, positive pairs are constructed from different augmentations of the same instance, while negative pairs are formed using views from distinct instances. In this paper, we introduce a noise generator that takes a molecule as input to produce chemically plausible variants, thereby generating new positive samples. The objective of contrastive learning is to encourage the model to recognize the shared chemical identity of a molecule by pulling its augmented views closer together in the latent space, simultaneously pushing apart the representations of different molecules. This is achieved by maximizing the similarity (e.g., cosine similarity) between the two views of the same molecule (positive pairs) while minimizing the similarity between views of different molecules within the same batch (negative pairs). Some works use multiple views from each instance by applying different transformations such as masking and dropping for molecules.^{10–12} Existing works primarily differ in how they construct these views:

- **Contrast based on 1D/2D/3D Information:** Early works perform contrastive learning on non-3D molecular representations. For example, SimSon¹³ constructs

positive pairs by enumerating different SMILES strings for the same molecule; MolCLR¹¹ generates different views by randomly augmenting the 2D molecular graph (e.g., masking atoms, deleting edges); and GROVER¹⁴ uses a dual-branch encoder to learn and align global graph and local motif features. For 3D information, 3DGCL¹⁵ constructs positive pairs from multiple low-energy conformers of the same molecule to preserve its chemical identity while introducing geometric variation.

- **Contrast based on Multimodal Information:** Some works learn by aligning molecular information of different modalities. A representative work, GraphMVP,¹² employs a dual-encoder system to process 2D and 3D graphs separately, aligning their representations through contrastive learning. Transformer-M¹⁶ uses a unified encoder to encode both 2D and 3D molecular graphs and then applies contrastive learning. MolSpectra¹⁷ pioneers the use of molecular spectra as a new view, contrasting it with the molecule's 3D conformation. MolMVC¹⁸ performs contrastive learning using information from 1D, 2D, and 3D perspectives. MolLM¹⁹ utilizes a language model to align textual information of molecules with their 2D and 3D representations for contrastive pretraining.

In contrast to the methods above, we unify contrastive learning and the denoising process at a high level. The positive pairs we construct are not derived from different modalities (e.g., 2D vs 3D) or simple graph augmentations, but consist of the original 3D conformation and a physically meaningful perturbed conformation generated by an adaptive noise generator. This design not only enhances the representation's robustness to noise, but more importantly, enables the learned representation to profoundly capture the dynamic geometric information that determines molecular properties by contrasting physically meaningful conformational pairs.

Denoising Self-Supervised Learning

Denoising methods (or denoising SSL) are a type of generative learning based on the fundamental idea of partially masking or adding noise to original data (e.g., atomic coordinates) and then training a model to recover the original information. When properly designed, such methods encourage encoders to substantiate the physical regularities of the Potential Energy Surface (PES) and benefit downstream property prediction.

- **Denoising based on 1D/2D Information:** At the SMILES string or 2D graph level, ChemBERTa²⁰ adopts the idea of BERT, pretrains by masking and predicting parts of the SMILES string. At the 2D graph level, HiMol²¹ performs masking and prediction at multiple levels—node, motif, and graph—to learn hierarchical molecular information. Similarly, MGSSL²² performs structural masking and prediction at both the atom and motif levels to learn multilevel representations.
- **Denoising based on 3D Coordinates:** This represents a trend. However, existing methods mostly adopt random, hand-crafted noise strategies. For example, Coord⁷ applies a random Gaussian noise to all atomic coordinates; SE(3)-DDM⁸ uses Gaussian noise to interatomic distances; and 3D-EMGP⁹ adopts a random multiscale Riemannian-Gaussian noise. To account for the anisotropy of the molecular force field, Frad⁵ employs a hybrid strategy combining dihedral angle noise and coordinate noise, which remains a random, nonadaptive

Table 1. Detailed Information of the MoleculeNet, QM7 and QM9 Benchmark Datasets

task-type	data set	#tasks	#compounds	split	metric	description
Classification	BBBP	1	2039	Scaffold	AUC-ROC	Blood-brain barrier penetration
	Tox21	12	7831	Random	AUC-ROC	Toxicity measurements on biological targets
	SIDER	27	1427	Random	AUC-ROC	Marketed drugs and their adverse drug reactions
	ClinTox	2	1484	Scaffold	AUC-ROC	Clinical trials for toxicity
	HIV	1	41127	Scaffold	AUC-ROC	Abilities to inhibit HIV replication
	BACE	1	1513	Scaffold	AUC-ROC	Binding affinity data for BACE-1 inhibitors
Regression	ESOL	1	1128	Random	RMSE	Water solubility
	FreeSolv	1	642	Random	RMSE	Hydration free energy of small molecules in water
	Lipophilicity	1	4200	Random	RMSE	Octanol/water distribution coefficient
	QM7	1	7160	Random	MAE	Electronic properties of small molecules from DFT calculations
	QM9	12	133,885	Random	MAE	Geometrical, energetic, electronic and thermodynamic properties of small organic molecules

scheme. A common point of these methods is that the added noise is predefined and cannot be adjusted according to the molecules' chemical structures.

This one-size-fits-all noise schedule mentioned above may produce implausible conformations, dilute informative training signals, and divert capacity toward correcting artifacts rather than modeling the true geometric–energetic landscape. To address this limitation, we introduce a learnable, adaptive noise generator that conditions on each atom's local environment to produce structure-aware, atom-specific perturbations. The generator is coupled with an equivariant denoising objective and energy-gap supervision to embed explicit physical priors, while a Kullback–Leibler (KL) regularization term prevents generator collapse and preserves perturbation diversity. This unified, energy-guided denoising framework provides physically meaningful self-supervision and complements the contrastive objective to learn robust 3D molecular representations.

RESULTS

We conduct extensive experimental evaluations to answer two major questions: (1) whether the proposed method is effective, or whether our energy-guided denoising pretraining can significantly improve molecular property prediction performance compared to pretraining based on random noise; and (2) whether our noise modeling mechanism can effectively learn the physical relationship between chemical structure and noise intensity.

Datasets

We utilize a large-scale molecular data set PCQM4Mv2 from Open Graph Benchmark (OGB)²³ as the pretraining data set. It includes 3.4 million equilibrium structures of molecules and their HOMO–LUMO gaps calculated using Density Functional Theory (DFT). The scale and diversity of PCQM4Mv2 make it highly suitable for pretraining, with the energy gap being used as an auxiliary task objective. For molecular property prediction, We validate our method on widely used molecular benchmark data sets including MoleculeNet,²⁴ QM7 and QM9,²⁵ which offer diversity in terms of scale, molecular types, and task types (classification and regression), as detailed in Table 1. For downstream tasks where 3D structures are not provided, we first generate 3D conformations from SMILES using RDKit, which

are then used as the model's input. Our data partitioning scheme follows previous studies.^{13,17}

Experimental Setup

The model is optimized using the AdamW optimizer, with a weight decay set to the order of 10^{-2} to prevent overfitting. The learning rate is scheduled using a cosine annealing scheduler, gradually decaying from an initial value of 10^{-4} . We also employ Exponential Moving Average (EMA) to maintain the exponential moving average of the encoder F_ϕ 's parameters, and use the EMA-updated parameters for validation and inference. The complete hyperparameter configurations for both pretraining and downstream fine-tuning are summarized in Table 2.

Regarding the hyperparameters of model architecture, the equivariant graph transformer encoder (F_ϕ) uses 20 layers of equivariant graph neural networks. The noise generator (G_θ) is a lightweight equivariant graph transformer network with 3 layers. LayerNorm is applied to the equivariant vector features for normalization. Pretraining with our method takes 4 days,

Table 2. Detailed Hyperparameters and Model Configurations Used for Pre-Training and Downstream Fine-Tuning^a

hyperparameter	pretraining	downstream fine-tuning
Optimizer	AdamW	AdamW
Warmup epochs	2	2
learning rate	1×10^{-4}	1×10^{-5}
Batch size	32	64
Total epochs	20	40
Weight decay	1×10^{-2}	1×10^{-2}
Dropout rate	0.2	0.2
Stochastic depth rate	0.05	0.1
Gradient clipping norm	100	100
EMA decay rate	0.999	0.999
Encoder layers	20	20
Generator layers	3	
LayerNorm on vector features	Yes	Yes
Cutoff radius (Å)	5.0	5.0
Max neighbors	50	50

^aAll downstream tasks share the same fine-tuning hyperparameters, selected via grid search on validation sets.

Table 3. Results of Classification Tasks, with Mean and Standard Deviation Reported^{a,b}

Model	BBBP	Tox21	SIDER	ClinTox	HIV	BACE
ChemBERTa ²⁰	88.04 (0.8590)	74.35 (4.8621)	60.02 (0.0157)	95.52 (1.4843)	69.46 (0.7709)	83.42 (0.0068)
GROVER-large ¹⁴	85.52 (3.9600)	74.26 (0.8603)		81.33 (9.5677)	70.17 (0.5801)	
GROVER-base ¹⁴	89.24 (1.2155)	70.94 (2.8930)		85.69 (0.4899)	72.62 (1.4382)	
MGSSL ²²	68.36 (0.9850)	76.60 (0.3950)	56.17 (0.0100)	67.27 (1.2019)	75.89 (1.1579)	82.74 (0.0118)
MolCLR ¹¹	81.83 (4.1910)	72.22 (3.3806)	58.23 (0.0951)	90.06 (1.8100)	78.54 (1.3828)	81.09 (0.0154)
GraphMVP ¹²	65.10 (0.0154)	81.78 (0.0037)	60.28 (0.0066)	80.78 (0.0208)	75.08 (0.0185)	69.62 (0.0625)
HiMol ²¹	86.21 (2.1866)	67.11 (0.7308)		59.32 (0.4636)	76.85 (0.9697)	
MolMVC ¹⁸	67.06 (0.0032)		56.81 (0.0126)	73.02 (0.1090)	67.51 (0.0076)	78.43 (0.0352)
MolLM ¹⁹	72.92 (0.0246)	79.34 (0.0020)		88.64 (0.0524)	69.05 (0.0518)	81.10 (0.0368)
GEM ²⁶	72.4 (0.4)	78.1(0.1)	67.2(0.4)	90.1(1.3)	80.6(0.9)	85.6(1.1)
CLAPS ²⁷	93.23 (1.846)	72.64(0.5932)	71.25(0.0638)	96.82(0.6824)	75.93(0.8395)	73.69(0.0739)
SimSon ¹³	92.30 (1.2124)	76.99 (0.2166)	63.93 (0.0227)	98.60 (0.7514)	76.65 (0.6226)	87.05 (0.0291)
Ours	94.26(0.2394)	85.73(0.8243)	67.94(0.3642)	98.21(0.7253)	78.37(0.5438)	88.36(0.0528)

^aThe best results are highlighted in bold. ‘-’ means data unavailable in the original paper. ^bThe results of all baselines are directly taken from their original papers.

Table 4. Results of Regression Tasks, Reporting Mean and Standard Deviation^{a,b}

Model	ESOL	FreeSolv	Lipophilicity	QM7
ChemBERTa ²⁰	0.7688 (0.0753)	1.9428 (0.3818)	0.7577 (0.0039)	175.16 (1.1176)
GROVER-large ¹⁴	0.7966 (0.1715)	1.7937 (0.3923)	0.9148 (0.0784)	
GROVER-base ¹⁴	0.8463 (0.0265)	2.0610 (0.1486)	0.8691 (0.0629)	
MolCLR ¹¹	0.7397 (0.0401)	1.5813 (0.0572)	0.6350 (0.0111)	84.33 (5.5607)
GraphMVP ¹²	0.7711 (0.0455)	1.0821 (0.0532)	0.6660 (0.0251)	
HiMol ²¹	0.7165 (0.0055)	1.3259 (0.0116)	0.6642 (0.0054)	
GEM ²⁶	0.798 (0.029)	1.877(0.094)	0.660(0.008)	58.9(0.8)
CLAPS ²⁷	0.5743 (0.0682)	0.9135(0.0384)	0.8264(0.1864)	79.53(3.5724)
SimSon ¹³	0.7052 (0.0322)	1.5151 (0.0833)	0.8391 (0.0204)	55.98 (0.143)
Ours	0.6582(0.0424)	0.8747(0.0847)	0.6224(0.0376)	52.22(0.2153)

^aThe best results are highlighted in bold. ‘-’ means data unavailable in the original paper. ^bRMSE is used as the evaluation metric for ESOL, FreeSolv, and Lipophilicity, while MAE is used for QM7. The results of all baselines are directly taken from their original papers.

Table 5. Performance (MAE) on QM9 Dataset^a

Model	μ (D)	α (a_0^3)	homo (meV)	lumo (meV)	gap (meV)	R^2 (a_0^3)	ZPVE (meV)	U_0 (meV)	U (meV)	H (meV)	G (meV)	C_v (cal/mol·K)
SchNet ²⁸	0.033	0.235	41.0	34.0	63.0	0.070	1.70	14.00	19.00	14.00	14.00	0.033
EGNN ²⁹	0.029	0.071	29.0	25.0	48.0	0.106	1.55	11.00	12.00	12.00	12.00	0.031
DimeNet++ ³¹	0.030	0.044	24.6	19.5	32.6	0.330	1.21	6.32	6.28	6.53	7.56	0.023
PaiNN ³⁰	0.012	0.045	27.6	20.4	45.7	0.070	1.28	5.85	5.83	5.98	7.35	0.024
SphereNet ³²	0.025	0.045	22.8	18.9	31.1	0.270	1.12	6.26	6.36	6.33	7.78	0.022
TorchMD-Net ³³	0.011	0.059	20.3	17.5	36.1	0.033	1.84	6.15	6.38	6.16	7.62	0.026
Ours (Training from scratch)	0.019	0.065	22.3	20.6	42.2	0.138	1.34	6.15	5.63	6.08	7.13	0.025
Transformer-M ¹⁶	0.037	0.041	17.5	16.2	27.4	0.075	1.18	9.37	9.41	9.39	9.63	0.022
SE(3)-DDM ⁸	0.015	0.046	23.5	19.5	40.2	0.122	1.31	6.92	6.99	7.09	7.65	0.024
3D-EMGP ⁹	0.015	0.046	23.5	19.5	40.2	0.122	1.31	6.92	6.99	7.09	7.65	0.024
Coord ⁷	0.016	0.052	17.7	14.7	31.8	0.450	1.71	6.57	6.11	6.45	6.91	0.020
Frad ⁵	0.010	0.037	15.3	13.7	27.8	0.342	1.42	5.33	5.62	5.55	6.19	0.020
MolSpectra ¹⁷	0.011	0.048	15.5	13.1	26.8	0.410	1.71	5.67	5.45	5.87	6.18	0.021
Ours	0.009	0.055	13.5	12.1	26.3	0.069	1.17	5.04	4.87	5.28	5.83	0.022

^aThe compared methods are divided into two groups: training from scratch and pre-training then fine-tuning. The best results are highlighted in bold.

corresponding to 187 GPU-hours, on a workstation equipped with 32 CPU cores, 512 GB memory, and two NVIDIA A100 80GB GPUs. The inference speed reaches 152.73 samples per second.

We employ three evaluation metrics: AUC-ROC, Root Mean Square Error (RMSE) and Mean Absolute Error (MAE), to

compare the performance of models on classification and regression tasks.

Results on MoleculeNet and QM7

The pretrained SE(3)-equivariant Graph Transformer F_ϕ and a lightweight prediction head were fine-tuned on the corresponding data set of each task. All experiments were repeated using three different random seeds, and we report the average

Table 6. Comparison of Prediction Performance on QM7, QM9 and Tox21 under Different Noise Strategies^a

Noise sampling scheme	QM9 (μ)	QM9 (α)	QM9 (homo)	QM9 (lumo)	QM9 (gap)	QM9 (R^2)	QM9 (ZPVE)	QM9 (U_0)	QM9 (U)	QM9 (H)	QM9 (G)	QM9 (C_v)	QM7	Tox21
Random	0.013	0.068	21.4	17.8	37.9	0.106	1.28	5.83	5.35	6.17	6.52	0.023	97.32	73.28
Ours	0.009	0.055	13.5	12.1	26.3	0.069	1.17	5.04	4.87	5.28	5.83	0.022	52.22	85.73

^aPerformance on QM7 and QM9 are reported in MAE, and performance on Tox21 is reported in AUC-ROC. The best results are highlighted in bold.

performance along with its standard deviation. Table 3 provides a detailed comparison of our method with other self-supervised pretraining methods for classification tasks.

The results in Table 3 clearly demonstrate the excellent performance and strong generalization of our method. Specifically, our method achieves state-of-the-art (SOTA) results on 3 data sets: BBBP, Tox21 and BACE. On the SIDER and ClinTox data sets, our method is also highly competitive. Concretely, on SIDER our method significantly outperforms most methods except for CLAPS. While on ClinTox, our method achieves 98.21, which is nearly on par with the best method SimSon (98.60), and our method exhibits lower variance, demonstrating the method's stability.

We further evaluate our method in 4 regression tasks. The results are in Table 4. Our method achieves SOTA performance on FreeSolv, Lipophilicity and QM7. In particular, on QM7 our method achieves the lowest MAE of 52.22 meV, outperforming SimSon. This may be attributed to energy gap prediction and energy-based denoising in our pretraining stage. On ESOL, although CLAPS performs best, our method achieves a highly competitive result of 0.6582, substantially surpassing most baseline methods such as MolCLR and SimSon.

Overall, our method shows SOTA performance in both classification and regression tasks. On the one hand, through contrastive learning, our model learns robust representations of molecules under perturbation, which is crucial for classification tasks. On the other hand, through energy-based denoising pretraining, our model can perceive fine-grained energy landscape states in physicochemical property regression tasks.

Results on QM9

On QM9, we fine-tune the pretrained model for each of the 12 target properties. Table 5 presents the results of our method and 12 existing methods. We categorize the compared methods into two groups: the first group includes 6 classic 3D graph neural networks trained from scratch on QM9, such as SchNet,²⁸ EGNN²⁹ and PaiNN;³⁰ the second group comprises 6 self-supervised pretraining methods such as Frad⁵ and MolSpectra.¹⁷

The results in Table 5 show that our method achieves SOTA performance on 8 of the 12 property prediction tasks. Notably, our method achieves the lowest error on all four energy-related properties (U_0 , U , H , and G). For example, our method has an MAE of 5.04 meV in predicting internal energy U_0 , which is over 5% lower than the best existing pretrained method Frad. These results show that our method, through energy-guided denoising pretraining, learns representations that lead to superior predictive performance on the molecular potential energy surface. Further considering the insights from our case studies, which show the model's ability to distinguish rigid and flexible regions, we can see that our energy-guided denoising pretraining can effectively capture chemically relevant information that is transferable to downstream tasks.

As for the remaining 4 tasks, our method also shows high competitiveness. Although TorchMD-Net excels on the electronic spatial extent (R^2), our method (0.069) still

significantly outperforms all the other pretrained methods. For zero-point vibrational energy (ZPVE), our result (1.17) is very close to SphereNet (1.12). For heat capacity (C_v) and polarizability (α), our performance is comparable to that of the best methods such as Transformer-M and Frad. To demonstrate the effectiveness of our proposed pretraining strategy, we compare the performance of our model under two settings: training from scratch on the downstream tasks and pretraining followed by fine-tuning. As shown in Table 5, the pretrained model consistently outperforms its from-scratch counterpart across all evaluation metrics. Since the same equivariant graph transformer backbone is used in both settings, the performance gain is attributed to the pretraining procedure rather than to architectural advantage.

In summary, the results on QM9, especially the advantages in predicting energy and electronic properties, validate the effectiveness of our method. It shows that by pretraining a physically more plausible energy-based denoising process, the model can predict quantum mechanics properties better.

Ablation Studies

Noise Paradigm. Our method assumes that for a denoising pretraining task to work, the perturbations should be physically plausible rather than random. To test this, we compare our adaptive noise generation strategy with random Gaussian noise (mean 0, standard deviation 0.5), which was used in the denoising pretraining method Coord.⁷

The results are presented in Table 6. In the homo- and lumo-property prediction tasks of the QM9 data set, simply replacing the random noise strategy with our adaptive one results in a significant improvement. For example, the MAE of the homo task decreases significantly from 21.4 to 13.5, reducing approximately 37%.

Being agnostic to molecular structures, random noise may produce perturbed structures with unrealistic bond lengths or atomic clashes. Consequently, the model would waste capacity on correcting these uninformative structure errors. In contrast, our adaptive noise generator perceives local chemical environments and produces challenging yet physically plausible perturbed structures. This helps the model learn high-quality physicochemical representations in low-energy states. Our pretraining with the adaptive noise generator requires 187 GPU-hours. Compared to the random noise strategy, the adaptive generator increases training time by approximately 23% and memory consumption by 17%, primarily due to the additional lightweight 3-layer equivariant network (the generator) and KL regularization.

Pretraining Proxy Tasks. Here we systematically evaluate each component of our pretraining framework through an ablation study, with the results detailed in Table 7.

The initial model, pretrained solely on the denoising task, performs poorly (homo error: 37.6). We attribute this failure to the collapse of the noise generator. Without appropriate constraint, the generator quickly learns a "shortcut" by

Table 7. Comparison Using Different Pre-Training Proxy Tasks

denoising	energy	CL	KL	QM9 (homo)	QM9 (lumo)	QM7	Tox21
✓				37.6	34.3	172.84	61.72
✓			✓	16.4	15.2	73.21	79.65
	✓			25.7	21.2	135.42	68.38
✓	✓		✓	13.9	12.7	57.95	83.80
✓	✓	✓	✓	13.5	12.1	52.22	85.73

producing near-zero noise. As a result, the model fails to learn meaningful representations in the denoising task.

To address this, we introduce KL divergence regularization to constrain the learned noise distribution to a prior. This effectively prevents generator collapse. As shown in Table 7, adding the KL loss substantially improves performance, reducing the homo error from 37.6 to 16.4.

Further performance gain is achieved by incorporating a self-supervised energy gap prediction task, stabilized by the same KL regularization. This task injects physical information into the representations, enhancing the model's understanding of the geometry-energy relationship and lowering the homo error to 13.9. This result highlights a significant improvement over using energy gap pretraining alone, which yields an error of 25.7.

Finally, our method achieves the best results by incorporating contrastive learning (CL) into the model. While the denoising and energy gap tasks enhance geometric and physical understanding, CL ensures representation robustness under perturbations.

Sensitive of the Prior ω . We pretrained models with different prior ω , and evaluated them on QM9 (HOMO, LUMO), QM7, and Tox21. The results are summarized in the Table 8: The results show that $\omega = 0.1$ yields the best

Table 8. Sensitivity of the KL prior ω on Downstream Performance

ω	QM9 (homo)	QM9 (lumo)	QM7	Tox21
0.01	26.8	23.4	156.35	63.57
0.1	13.5	12.1	52.22	85.23
1.0	22.7	19.4	122.74	69.82

performance across all tasks. When $\omega = 0.01$, the KL constraint is too weak to prevent generator collapse, leading to near-zero noise and poor representation learning. When $\omega = 1.0$, the regularization overly restricts the noise scale, limiting the diversity of perturbed structures and impairing downstream performance.

Case Study

To complement our method with more intuitive and interpretable insights, we use four molecules in case studies to visualize the relationship between adaptive noise and local chemical structure. Furthermore, we explore whether the generator has learned the chemistry and energy information.

Noise Visualization Analysis. We feed some molecules into the pretrained noise generator and visualize the predicted noise for each atom, as shown in Figure 1. Deeper color indicates that the model applies stronger perturbation to the corresponding atom.

From Figure 1, we can observe that the noise allocation is nonuniform. High-noise regions involve the acetyl group. For example, in Figure 1a, the methyl carbon (Atom 0, Noise =

0.2665), the carbonyl oxygen (Atom 2, Noise = 0.2608), and the carbon-oxygen atoms of the ester bond (Atom 1, Atom 3) all obtain large noise. From the chemical perspective, the acetyl group is connected to the benzene ring by a single bond and possesses high conformational flexibility, especially the terminal methyl ($-\text{CH}_3$) group, which can rotate freely. Applying larger shift perturbations to these atoms is less likely to disrupt the molecule's fundamental scaffold. Therefore, the energy change is not significant. While the acetyl oxygen (Atom 2) receives high noise, the carboxyl oxygen (Atom 11, Noise = 0.2214) in the acid group ($-\text{COOH}$) is perturbed to a smaller extent. The difference is not captured by a simplistic view of bond rotatability alone. The carboxyl group is a resonance-stabilized, planar unit where the π -system is delocalized over the $\text{O}=\text{C}-\text{O}$ moiety. This electronic structure imposes significant rigidity, making large, out-of-plane displacements of its heavy atoms energetically costly. Conversely, low-noise regions are mainly comprised of the carbon atoms of the benzene ring (e.g., Atom 8, Noise = 0.1817; Atom 5, Noise = 0.1871), which form the core scaffold of the aspirin. The benzene ring is a highly rigid, planar conjugated system with high structural stability. Applying a large shift to any atom on the ring could severely disrupt its conjugated structure and planarity, leading to an increase in molecular energy and forming a physically "unreasonable" structure.

For Mitotane (Figure 1b), a molecule featuring two chlorinated aromatic rings connected by a central quaternary carbon, the lowest noise values are observed on the benzene ring carbons, preserving the rigid aromatic scaffold. In contrast, the dichloromethyl group ($-\text{CHCl}_2$) exhibits substantially higher noise (Atom 1, Noise = 0.2490), reflecting the conformational freedom of this flexible substituent.

For Solriamfetol (Figure 1d), a phenylalanine derivative with a carbamate group, the aromatic ring again displays uniformly low noise (Atom 8–13, average Noise = 0.1844). The benzylic methylene (Atom 7, Noise = 0.2429) and the chiral methine carbon (Atom 1, Noise = 0.2366) receive elevated noise, consistent with their rotational freedom.

For Oxybenzone (Figure 1c), a common UV filter containing a benzophenone core with methoxy and hydroxy substituents. Both aromatic rings exhibit characteristically low noise. In the methoxy group, the ether oxygen (Atom 1, Noise = 0.2025) and especially the terminal methyl (Atom 0, Noise = 0.2145) receive high noise, capturing the free rotation of the $\text{O}-\text{CH}_3$ bond.

The above cases demonstrate that the noise generator preferentially assigns higher noise to atoms located in conformationally flexible regions (e.g., terminal methyl groups, ether linkages, freely rotating substituents). In comparison, it allocates lower noise to rigid structural scaffolds (aromatic rings, conjugated systems, resonance-stabilized moieties).

To check whether these cases are a spurious effect, we extend our analysis to the entire QM9. Specifically, we categorize all carbon atoms based on their hybridization states (sp , sp^2 , sp^3 , ring/aromatic) and compute the average noise assigned by our generator to each category. As shown in Figure 2, the results reveal a clear and physically meaningful hierarchy. sp^3 -hybridized carbons (e.g., methyl and methylene groups) exhibit the highest average noise. The single bonds in sp^3 centers allow rotation and bending with relatively low energetic cost, making larger perturbations physically plausible. sp^2 -hybridized carbons in nonaromatic environments (e.g., olefinic or carbonyl carbons) show intermediate noise levels. In contrast, sp^2 -

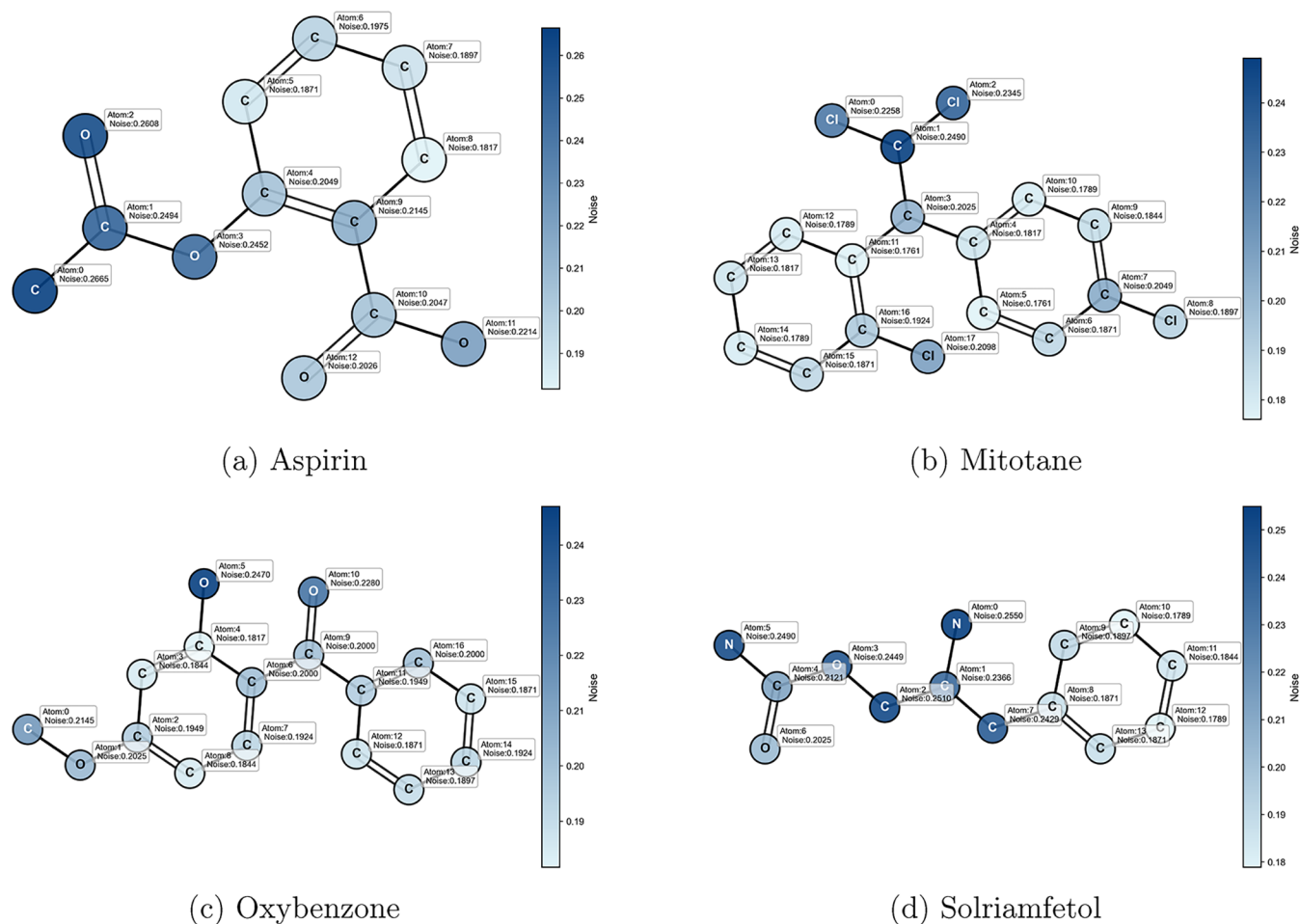


Figure 1. Visualization of noise generated by the noise generator for four molecules.

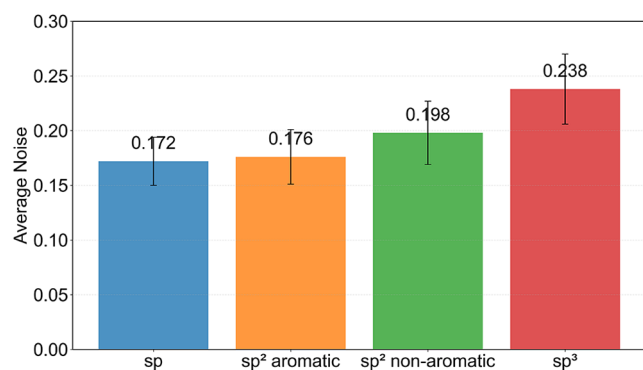


Figure 2. Average noise assigned by the generator to carbon atoms with different hybridization states across the QM9. Error bars indicate standard deviations.

hybridized carbons within aromatic rings display lower noise, capturing the stabilization.

Perturbation and Energy Response. We further compare the energy changes in the molecules before and after applying noise for each atom by different noise strategies: our adaptive noise and the random noise (mean = 0, standard deviation = 0.5). In Figure 3, we sequentially perturb the atoms in molecules and use sMAPE (symmetric Mean Absolute Percentage Error)³⁴ to measure the change in total molecular energy evaluated by RDKit.³⁵ sMAPE measures the relative energy change, enabling fair comparisons across atoms with different baseline energies.

To establish an intuitive correspondence between energy sensitivity and noise magnitude, we further transform sMAPE into 1/sMAPE: a higher 1/sMAPE value indicates a smaller relative energy change, i.e., the atom is less sensitive to perturbation.

As shown in Aspirin (Figure 3a), the largest perturbation is applied to the highly flexible methyl carbon (Atom 0), yet the resulting energy change remains moderate, indicating that this perturbation is energetically plausible. In contrast, the yellow bars (random noise) show substantial energy increases when random noise is applied. And the atoms on the benzene ring exhibit greater energy changes than those in other regions, both before and after noise is added.

In Mitotane (Figure 3b), the flexible dichloromethyl group (Atom 1) receives the largest adaptive perturbation. Despite this, the resulting energy change remains modest, indicating that the distortion may stay within a low-energy region of the potential energy surface. For Oxybenzone (Figure 3c), the freely rotating terminal methyl (Atom 0) and the ether oxygen (Atom 1) again show large adaptive perturbations with small energy penalties. Similarly, in Solriamfetol (Figure 3d), the benzylic methylene (Atom 7) demonstrates high adaptive noise at low energetic cost.

In summary, our model effectively explores a richer conformational space while avoiding high-energy, unstable regions, which results in more consistent and stable learning signals during pretraining.

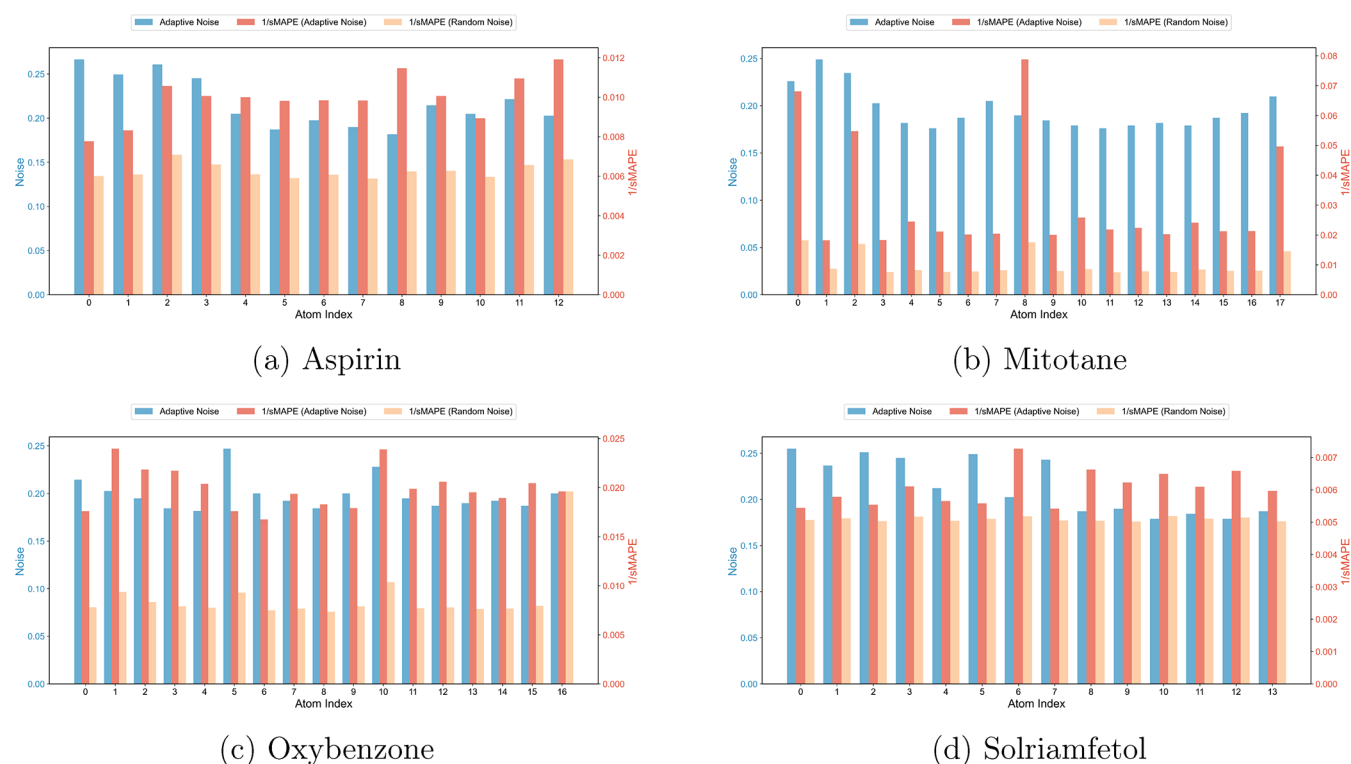


Figure 3. Comparison of molecular energy changes resulting from adaptive noise and random noise. The blue bars represent the case that the adaptive noise is applied per atom, while the red and yellow bars show the 1/sMAPE after applying adaptive and random noise, respectively. A higher 1/sMAPE value corresponds to a smaller molecular energy change.

CONCLUSION AND DISCUSSION

This paper introduces an energy-guided denoising contrastive learning pretraining paradigm for molecular property prediction. The innovation lies in an adaptive noise generator rather than conventional random noise strategies to generate noise. By integrating denoising, energy supervision and contrastive learning, our method promotes the model to learn the molecular potential energy surface (PES), achieving SOTA performance across multiple benchmarks.

Our method successfully overcomes several core technical challenges in 3D molecular representation learning. First, we designed a learnable noise generator to resolve the fundamental limitation of “one-size-fits-all” random noise, which ignores the heterogeneous chemical environments within a molecule and generates physically implausible training samples. Second, we used a KL divergence regularization term to mitigate the “generator collapse” problem, where the noise generator might converge to a trivial solution of zero noise. Finally, our model simultaneously learns the molecular force field and the intrinsic link between geometry and energy through a synergistic multitask objective pretraining. As demonstrated in the case studies, our model intuitively distinguishes between the rigid benzene ring and the flexible acetyl group, a “chemical intuition” unattainable by previous studies.

Despite its success, our method still has limitations worthy of further investigation. First, the model’s generalization is dependent on the scale and accuracy of the pretraining data set. Future work could explore pretraining on more diverse and higher-quality quantum chemistry data sets. Second, the current paradigm is based on equilibrium structures, which may not capture the large-scale structural changes involved in many biochemical processes. Extending the framework to learn from

molecular dynamics trajectories is a promising, albeit challenging direction.

METHODS

Problem Definition and Framework

We begin by formally defining the problem. A molecule $\mathcal{M}$ is represented as a set of its atoms and a set of their corresponding 3D coordinates, i.e., $\mathcal{M} = (\mathcal{A}, \mathcal{R})$. Here, $\mathcal{A} = \{a_1, a_2, \dots, a_N\}$ is the set of N atoms in the molecule, typically represented by atomic numbers, $a_i \in \{1, \dots, 118\}$. $\mathcal{R} = \{\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N\} \in \mathbb{R}^{N \times 3}$ is the set of 3D coordinate vectors of these atoms in $\mathcal{A}$.

Our goal is to learn the representations of molecules that effectively depict the molecular energy landscape and possess strong generalization capabilities, with which to accurately predict the properties of the molecules.

To tackle the problem above, we propose an Energy-guided Denoising Contrastive Learning (EDCL) framework that consists of a pretraining phase and a fine-tuning phase, as shown in Figure 4. During pretraining (Figure 4(a)), we start from a ground truth molecular structure and deliberately perturb it by adding controlled noise, generating a perturbed structure. Both the ground truth and perturbed structures are then encoded by an equivariant graph Transformer to obtain their latent representations. These representations are then processed by dedicated prediction heads: a denoising prediction head that aims to reconstruct the original structure from its perturbed structure, and an energy prediction head that estimates the energy gap from the ground truth structure. Meanwhile, a contrastive learning objective is applied to pull the representations of the ground truth and perturbed views close, enhancing their semantic invariance. During fine-tuning (Figure 4(b)), the pretrained equivariant Transformer encoder is transferred and fine-tuned on downstream tasks, with task-specific heads adapted to predicting target molecular properties. In our pretraining and fine-tuning pipeline, only heavy atoms are considered.

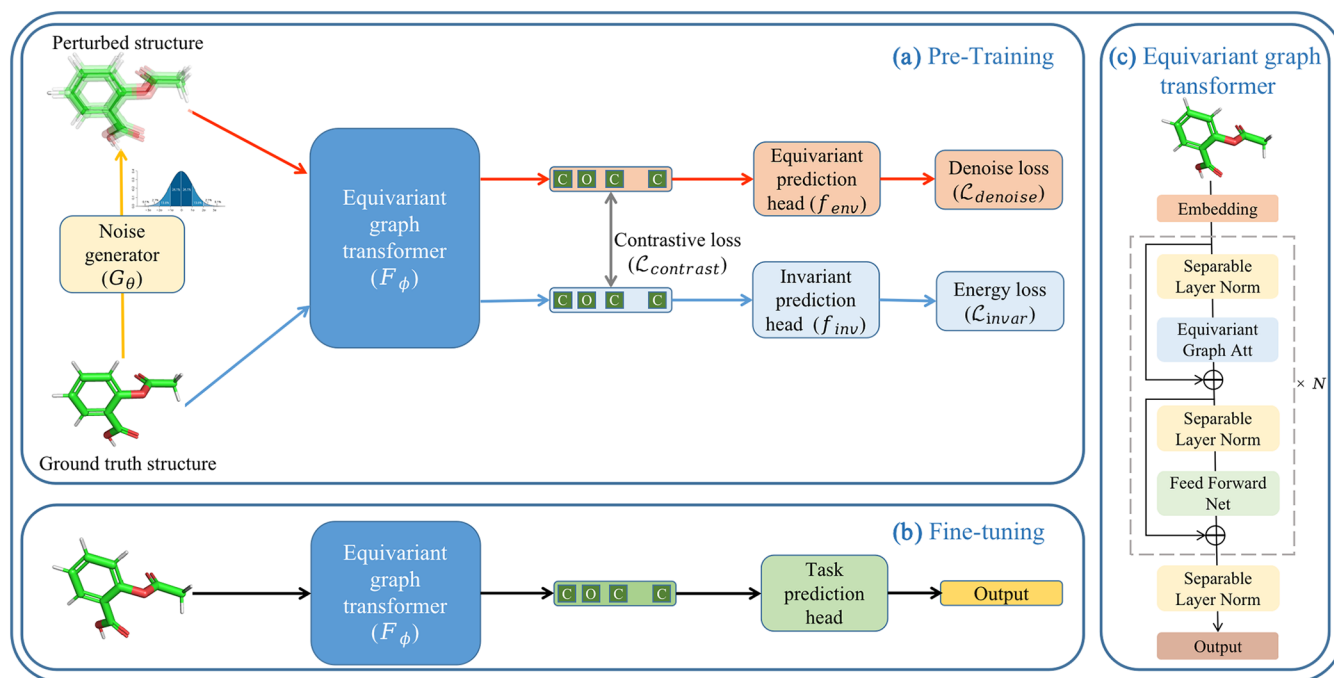


Figure 4. Framework of Energy-guided Denoising Contrastive Learning for Molecular Property Prediction. (a) **Pretraining phase.** The input is a ground truth structure, it is fed into a noise generator to produce noise, which is added to the ground truth structure to obtain a perturbed structure. Both the ground truth structure and perturbed structure are then passed through an equivariant graph Transformer for feature encoding. Subsequently, these encoded features are fed into corresponding prediction heads, and relevant losses are computed. During this process, embeddings from both ground truth and perturbed structures undergo contrastive learning loss calculation. (b) **Fine-tuning phase.** The equivariant graph Transformer encoder learned during pretraining is applied to downstream tasks. Task-specific fine-tuning is performed according to the particular objective of each target task. (c) **The equivariant graph transformer architecture,** which consists of three major modules: (1) Embedding: Atom embedding and edge embedding. (2) Graph-based equivariant module, composed of Equivariant graph attention for message passing, FFN for feature transformation using equivariant linear layers, and separable layerNorms for independent normalization of scalar/vector features. (3) Output layer: performing equivariant shift vector prediction.

Figure 4(c) is the architecture of our equivariant graph transformer, which is composed of three major modules: an embedding layer, a graph-based equivariant network, and an output layer. The embedding layer is responsible for generating initial embedding for atoms and edges. The graph-based equivariant network is based on the Equiformer V2 framework,³⁶ which is a Transformer-like framework with three E3-equivariant variation components: (i) Equivariant Graph Attention, (ii) Feed-Forward Network (FFN), and (iii) Separable Layer Normalization.³⁷ In the Equivariant Graph Attention module, equivariant and invariant features are processed separately. The final equivariant features are then passed to the output layer (equivariant prediction head) to predict the atom-wise vectors via an equivariant linear layer. To predict the molecular scale value of energy, we use a feed-forward network to transform irreps features on each node into a scalar and then sum over all nodes.

Pretraining Stage: Energy-Guided Denoising Contrastive Learning

Pretraining Pipeline. In this stage, as illustrated in Figure 4 (a), our goal is to learn an SE(3)-equivariant graph transformer encoder, F_ϕ , with chemical and physical knowledge of molecules using a large-scale molecule data set.

We perform physically plausible perturbation sampling within the low-energy neighborhood of the molecular potential energy surface (PES) using a learnable noise generator G_θ , which generates a perturbed structure $\tilde{\mathcal{M}} = (\mathcal{A}, \tilde{\mathcal{R}})$ from the ground truth molecule $\mathcal{M}$. The SE(3)-equivariant graph transformer encoder and noise generator are based on the Equiformer V2 framework.³⁶ Then, the encoder F_ϕ learns its parameters ϕ by optimizing a multitask objective, which includes

- **The equivariant denoising task:** Predicting the atom-level shift vectors to restore the ground truth structure $\mathcal{M}$ from the perturbed structure $\tilde{\mathcal{M}}$.
- **The contrastive alignment task:** Pulling the pair $(\mathcal{M}, \tilde{\mathcal{M}})$ close in the representation space, while pushing the others apart as negative samples.
- **The energy gap supervision prediction task:** Predicting the energy gap y_{energy} via DFT,³⁸ from the representation of the ground truth structure $\mathcal{M}$. The energy gap is one of the most fundamental quantum chemical properties of a molecule, as it is related to reactivity, photoexcitation, and charge transport.

In such a way, the encoder F_ϕ can capture the structural and energetic information as well as the potential correlation between them. Below, we elaborate on each step of the pretraining process in detail.

Generating Perturbed Structures. We construct a learnable noise generator, G_θ , with the ground truth structure $\mathcal{M}$ as input to output a predicted noise σ_i for atom i . Then, we generate the perturbed coordinates via reparameterization sampling as eq 1

$$\tilde{\mathbf{r}}_i = \mathbf{r}_i + \sigma_i \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (1)$$

where $\mathbf{r}_i$ is the coordinate of atom i . The perturbed structure is $\tilde{\mathcal{M}} = (\mathcal{A}, \tilde{\mathcal{R}})$ where $\tilde{\mathcal{R}} = \{\tilde{\mathbf{r}}_i\}_{i=1}^N$. To avoid implausible structures, we employ validity check to reject samples with $\min_{(i,j)} \|\tilde{\mathbf{r}}_i - \tilde{\mathbf{r}}_j\|_2 < r_{\min}$. Here, we set $r_{\min} = 0.8$.

Learning Molecular Representations. A shared-parameter SE(3)-equivariant Graph Transformer neural network encoder, F_ϕ , is used to encode $\mathcal{M}$ and $\tilde{\mathcal{M}}$ respectively. The atom-level representation of molecule $\mathcal{M} = (\mathcal{A}, \mathcal{R})$ is denoted as $H = F_\phi(\mathcal{M}) \in \mathbb{R}^{N \times d}$ where N is the number of atoms and d is the feature dimension.

The raw 3D molecular structure is encoded into input representations as summarized in Table 9. Specifically, the atomic

Table 9. Initial Encoding of 3D Molecular Structure in Equivariant Graph Transformer

feature type	input data	encoding method/architecture
Node Features	Atomic Number (Z_i)	Learnable Embedding Lookup Table
Edge Distance	Interatomic distance (r_{ij})	Gaussian RBF + 2-layer MLP (LN & SiLU)
Edge Geometry	Unit vector ($\hat{r}_{ij}$)	Spherical Harmonics up to degree $L_{\max}$ ($Y(\hat{r}_{ij})$)
Coordinate Transformation	Relative position ($\tilde{r}_{ij}$)	Calculation of Wigner-D rotation matrices D_{ij} based on $\hat{r}_{ij}$
Chemical Context	Source & target atomic numbers (Z_i, Z_j)	Concatenated with RBF and passed through the Radial MLP

number Z_i of each atom is mapped to a learnable embedding to produce node features. For each edge between atom i and atom j , the interatomic distance r_{ij} is expanded using Gaussian Radial Basis Functions (RBF), and the resulting expansion is passed through a two-layer MLP with Layer Normalization (LN) and Sigmoid-Weighted Layer Normalization (SiLU) activation to obtain radial features. The unit direction vector $\hat{r}_{ij}$ is embedded via spherical harmonics $Y(\hat{r}_{ij})$ up to a maximum degree $L_{\max}$, yielding angular irreducible representations (irreps) spanning degrees from 0 to $L_{\max}$. To maintain equivariance, the relative position $\tilde{r}_{ij}$ is also used to compute the Wigner-D rotation matrices D_{ij} , which align edge features to a reference coordinate frame. Finally, the atomic numbers of both the source and target atoms (Z_i and Z_j) are concatenated with the RBF features and passed through the radial MLP, generating chemically informed edge features. More details can refer to the Equiformer V2.³⁶

The molecule-level representation is denoted by $H_{\text{graph}} = \text{mean}(H) \in \mathbb{R}^d$. Similarly, we can obtain the atom-level and graph-level representations $\tilde{H}, \tilde{H}_{\text{graph}}$ of the perturbed structure. These are then used in the following steps:

- **Contrastive learning:** In the molecule-level feature space, H_{graph} and $\tilde{H}_{\text{graph}}$ form a positive pair, while the other combinations in this batch are treated as negative samples.
- **Equivariant noise predicting:** An equivariant prediction head, denoted as f_{env} , is employed to predict atom-level shifts for restoring the original molecular structure from its perturbed structure (eq 2).

$$\hat{\Delta} = f_{\text{env}}(\tilde{H}) \in \mathbb{R}^{N \times 3} \quad (2)$$

- **Invariant energy predicting:** Since the energy gap is a global scalar that should be invariant to translation and rotation, an invariant prediction head f_{inv} is used to predict the HOMO-LUMO energy gap as a supervisory signal to guide the denoising direction (eq 3).

$$\hat{y} = \text{mean}(f_{\text{inv}}(H)) \in \mathbb{R},$$

where $f_{\text{inv}}(H) = \{e_1, \dots, e_N\} \in \mathbb{R}^N$ (3)

Optimization Objectives. Our method consists of three optimization objectives: Contrastive Loss, Denoise Loss and Energy Loss. Below, we introduce these loss functions.

Contrastive Loss. For a batch of molecular pairs $\{(\tilde{M}_b, M_b)\}_{b=1}^B$, we employ the InfoNCE loss.³⁹ For notational convenience, in eq 4, we replace H_{graph} and $\tilde{H}_{\text{graph}}$ with g and $\tilde{g}$ respectively

$$\mathcal{L}_{\text{contrast}} = -\frac{1}{B} \sum_{b=1}^B \log \frac{\exp(\text{sim}(\tilde{g}_b, g_b)/\tau)}{\sum_{b'=1}^B \exp(\text{sim}(\tilde{g}_b, g_{b'})/\tau)} \quad (4)$$

where $\text{sim}()$ is the cosine similarity function and τ is the temperature coefficient. The contrastive learning loss explicitly aligns the representations of the perturbed structures and the corresponding ground truth structures to enhance robustness against perturbation.

Denoising Loss. Inspired by physics and generative model theory, denoising constitutes the core pretraining task. From the physical perspective, the distribution of molecular structures at equilibrium follows the Boltzmann distribution (eq 5)⁷

$$p_{\text{phys}}(\mathcal{R}) = \frac{1}{Z} \exp\left(-\frac{E(\mathcal{R})}{kT}\right) \quad (5)$$

where $E(\mathcal{R})$ is the molecular potential energy function. k , T , Z are the Boltzmann constant, temperature and partition function, respectively. Taking the logarithm and then the gradient as eq 6

$$\nabla_{\mathcal{R}} \log p_{\text{phys}}(\mathcal{R}) = -\frac{1}{kT} \nabla_{\mathcal{R}} E(\mathcal{R}) \quad (6)$$

Here, ∇ represents the gradient operator. This equation shows that the score function (gradient) of the data distribution (structure) is proportional to the molecular force field (negative potential energy gradient). Therefore, learning the score function of the structure distribution is physically equivalent to learning an implicit molecular force field.

From the generative modeling perspective, Denoising Score Matching (DSM)⁴⁰ predicts how to recover the original structure $\mathcal{R}$ from a noise-perturbed structure $\tilde{\mathcal{R}}$ to learn the score function $\nabla_{\mathcal{R}} \log p_{\text{data}}(\mathcal{R})$ of the data distribution $\mathcal{R} \sim p_{\text{data}}$. For an isotropic Gaussian distribution $\tilde{\mathcal{R}} \sim \mathcal{N}(\mathcal{R}, \sigma^2 I)$, DSM is equivalent to a denoising regression (eq 7)⁷

$$\mathbb{E}_{p(\tilde{\mathcal{R}})} \|F_{\Psi}(\tilde{\mathcal{R}}) - \nabla_{\tilde{\mathcal{R}}} \log p(\tilde{\mathcal{R}})\|^2 \propto \mathbb{E}_{p(\tilde{\mathcal{R}})} \left\| F_{\Psi}(\tilde{\mathcal{R}}) - \frac{\mathcal{R} - \tilde{\mathcal{R}}}{\sigma^2} \right\|^2 \quad (7)$$

where F_{Ψ} is the model to be learned. (Equation 7) statistically bridges denoising (right side) and score matching (left side).

In our work, the isotropic Gaussian noise is generated by a parametrized noise generator, and then the noise is applied to each atom (eq 8)

$$q(\tilde{\mathcal{R}}|\mathcal{R}) = \prod_{i=1}^N \mathcal{N}(\tilde{\mathbf{r}}_i; \mathbf{r}_i, \sigma_i^2 \mathbf{I}_3) \quad (8)$$

For this Gaussian noise model, the conditional score function has a simple form as eq 9

$$\nabla_{\tilde{\mathbf{r}}_i} \log q(\tilde{\mathcal{R}}|\mathcal{R}) = -\frac{\tilde{\mathbf{r}}_i - \mathbf{r}_i}{\sigma_i^2} = \frac{\Delta_i}{\sigma_i^2} \quad (9)$$

Where $\Delta_i = \tilde{\mathbf{r}}_i - \mathbf{r}_i \in \mathbb{R}^3$ is the shift vector of atom i between the perturbed and ground-true structures, serving as the training target. The shift vector Δ_i is translation-invariant and rotation-equivariant, i.e., $\Delta_i \rightarrow Q\Delta_i$ for $Q \in SO(3)$. We therefore predict shifts with an SE(3)-equivariant head f_{env} (eq 2) from equivariant features $\tilde{H}_i$. Compared to directly predicting absolute coordinates, predicting shift vectors significantly improves training stability and geometric consistency.

Specifically, transforming the output $\hat{\Delta}_i$ of our denoising head as $\hat{s}_i \equiv \hat{\Delta}_i/\sigma_i^2$. Therefore, based on (eq 7), the denoising loss function $\mathcal{L}_{\text{denoise_nll}}$ is as eq 10

$$\begin{aligned} \mathcal{L}_{\text{denoise_nll}} &= \mathbb{E}_{(\tilde{\mathcal{R}}, \mathcal{R})} \sum_{i=1}^N \frac{1}{\sigma_i^2} \left\| \hat{\Delta}_i + \sigma_i^2 \cdot \nabla_{\tilde{\mathbf{r}}_i} \log q(\tilde{\mathcal{R}}|\mathcal{R}) \right\|^2 \\ &= \mathbb{E}_{(\tilde{\mathcal{R}}, \mathcal{R})} \sum_{i=1}^N \frac{1}{\sigma_i^2} \left\| -\sigma_i^2 \hat{s}_i + \sigma_i^2 \cdot \nabla_{\tilde{\mathbf{r}}_i} \log q(\tilde{\mathcal{R}}|\mathcal{R}) \right\|^2 \\ &= \mathbb{E}_{(\tilde{\mathcal{R}}, \mathcal{R})} \sum_{i=1}^N \sigma_i^2 \left\| \hat{s}_i - \nabla_{\tilde{\mathbf{r}}_i} \log q(\tilde{\mathcal{R}}|\mathcal{R}) \right\|^2 \end{aligned} \quad (10)$$

It is clear that minimizing the denoising loss $\mathcal{L}_{\text{denoise_nll}}$ is essentially minimizing the weighted mean squared error between $\hat{s}_i$ and the ground-truth score $\nabla_{\mathbf{r}_i} \log q(\tilde{\mathbf{R}}|\mathcal{R})$. σ_i^2 acts as an adaptive weight, ensuring consistent objective scaling across different noise levels.

However, when minimizing $\mathcal{L}_{\text{denoise_nll}}$, the noise scale σ_i may degenerate to a trivial solution that $\sigma_i \rightarrow 0$. As we know, molecules are not static in 3D space but vibrate around the equilibrium structure on the potential energy surface (PES). A reasonable perturbation structure should lie within the low-energy neighborhood of the ground truth equilibrium structure. It generally holds that like (eq 11)

$$E(\mathcal{R} + \delta) \approx E(\mathcal{R}) \text{ for small perturbation } \|\delta\| \quad (11)$$

To prevent noise scale degeneracy, we use a KL regularization term (eq 12) to encourage the learned noise distribution to approximate a reasonable prior distribution

$$\mathcal{L}_{\text{KL}} = \frac{1}{N} \sum_{i=1}^N D_{\text{KL}}(\mathcal{N}(0, \sigma_i^2) \parallel \mathcal{N}(0, \omega^2)) \quad (12)$$

where ω is the prior standard deviation, which is set to 0.1. On the one hand, $\mathcal{L}_{\text{KL}}$ prevents the trivial solution $\sigma_i \rightarrow 0$, and on the other hand, it prevents excessive growth of σ_i .

Finally, the joint optimization objective (eq 13) is a weighted sum of the denoising loss and the KL regularization term

$$\mathcal{L}_{\text{denoise}} = \mathcal{L}_{\text{denoise_nll}} + \lambda \mathcal{L}_{\text{KL}} \quad (13)$$

where λ is a weight coefficient, which is set to 1. The model can simultaneously learn the representations of molecular structures and robustness to reasonable perturbations by optimizing this objective.

Energy Loss. We design an SE(3)-invariant prediction head (eq 3) to predict the energy gap of molecules, then compute the energy loss (eq 14) between the predicted energy gap $\hat{y}$ and the target energy gap y_{energy}

$$\mathcal{L}_{\text{invar}} = (\hat{y} - y_{\text{energy}})^2 \quad (14)$$

where y_{energy} is the ground truth energy gap label. The energy supervision signal only works on the true structure branch. This design prevents the noise on the perturbed branch from directly propagating to the energy loss, thus ensuring the stability of the energy prediction task.

Total Loss. Finally, the total loss (eq 15) is defined as the weighted sum of the above losses

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{denoise}} + \alpha \mathcal{L}_{\text{contrast}} + \beta \mathcal{L}_{\text{invar}} \quad (15)$$

where α and β are hyperparameters used to balance the respective tasks. Here, we set α and β to 0.1 and 10, respectively. The hyperparameters α , β and λ are to balance the three loss components, ensuring that no single task dominates the optimization. Preliminary experiments show that when λ is set to 1, the model can effectively avoid the collapse problem. Then, by performing a grid search over a reasonable range ($\alpha \in [0.01, 1]$, $\beta \in [1, 100]$), we select the combination that gets the best validation performance on PCQM4Mv2.

Fine-tuning Stage: Property Prediction for Downstream Tasks

In the fine-tuning stage (Figure 4 (b)), only the equivariant Graph Transformer encoder F_{ϕ} is retained as the feature extractor for various specific downstream molecular property prediction tasks.

For any given downstream task, the molecular representations from the pretrained encoder F_{ϕ} are fed into a lightweight, task-specific prediction head P_{task} , implemented by a randomly initialized multilayer perceptron (MLP). The entire model is then fine-tuned end-to-end to predict molecular properties.

■ ASSOCIATED CONTENT

Data Availability Statement

The data sets used for training and testing are available at <https://moleculenet.org/> and <https://ogb.stanford.edu/>.

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) Adelusi, T. I.; Oyedele, A.-Q. K.; Boyenle, I. D.; Ogunlana, A. T.; Adeyemi, R. O.; Ukachi, C. D.; Idris, M. O.; Olaoba, O. T.; Adedotun, I. O.; Kolawole, O. E.; et al. Molecular modeling in drug discovery. *Inf. Med. Unlocked* **2022**, 29, No. 100880.
- (2) Han, J.; Rong, Y.; Xu, T.; Huang, W. Geometrically Equivariant Graph Neural Networks: A Survey. 2022, arXiv:2202.07230. arXiv.org e-Printarchive. <https://arxiv.org/abs/2202.07230>.
- (3) Chen, H.; Wang, Y.; Guo, T.; Xu, C.; Deng, Y.; Liu, Z.; Ma, S.; Xu, C.; Xu, C.; Gao, W. In *Pre-Trained Image Processing Transformer*; Proceedings of the IEEE/CVF conference on computer vision and pattern recognition IEEE, 2021; pp 12299–12310 DOI: 10.1109/cvpr46437.2021.01212.
- (4) Qiu, X.; Sun, T.; Xu, Y.; Shao, Y.; Dai, N.; Huang, X. Pre-trained models for natural language processing: A survey. *Sci. China: Technol. Sci.* **2020**, 63, 1872–1897.
- (5) Feng, S.; Ni, Y.; Lan, Y.; Ma, Z.-M.; Ma, W.-Y. In *Fractional Denoising for 3D Molecular Pre-Training*; Proceedings of the 40th International Conference on Machine Learning, PMLR, 2023; pp 9938–9961.
- (6) Song, Y.; Ermon, S. In *Generative Modeling by Estimating Gradients of the Data Distribution*; Advances in Neural Information Processing Systems 32, NIPS, 2019.
- (7) Zaidi, S.; Schaarschmidt, M.; Martens, J.; Kim, H.; Teh, Y. W.; Sanchez-Gonzalez, A.; Battaglia, P.; Pascanu, R.; Godwin, J. Pre-Training via Denoising for Molecular Property Prediction. 2022, arXiv:2206.00133. arXiv.org e-Printarchive. <https://arxiv.org/abs/2206.00133>.
- (8) Liu, S.; Guo, H.; Tang, J. Molecular Geometry Pretraining with se (3)-Invariant Denoising Distance Matching. 2022, arXiv:2206.13602. arXiv.org e-Printarchive. <https://arxiv.org/abs/2206.13602>.
- (9) Jiao, R.; Han, J.; Huang, W.; Rong, Y.; Liu, Y. Energy-motivated equivariant pretraining for 3d molecular graphs. *Proc. AAAI Conf. Artif. Intell.* **2023**, 37, 8096–8104.

- (10) Wang, J.; Guan, J.; Zhou, S. Molecular Property Prediction by Contrastive Learning with Attention-Guided Positive Sample Selection. *Bioinformatics* **2023**, *39*, No. btad258.
- (11) Wang, Y.; Wang, J.; Cao, Z.; Barati Farimani, A. Molecular contrastive learning of representations via graph neural networks. *Nat. Mach. Intell.* **2022**, *4*, 279–287.
- (12) Liu, S.; Wang, H.; Liu, W.; Lasenby, J.; Guo, H.; Tang, J. Pre-Training Molecular Graph Representation with 3D Geometry. 2021, arXiv:2110.07728. arXiv.org e-Printarchive. <https://arxiv.org/abs/2110.07728>.
- (13) Lee, C. E.; Kim, J. S.; Min, J. H.; Han, S. W. SimSon: simple contrastive learning of SMILES for molecular property prediction. *Bioinformatics* **2025**, *41*, No. btaf275.
- (14) Rong, Y.; Bian, Y.; Xu, T.; Xie, W.; Wei, Y.; Huang, W.; Huang, J. In *Self-Supervised Graph Transformer on Large-Scale Molecular Data*; Advances in Neural Information Processing Systems 33 NIPS, 2020; pp 12559–12571.
- (15) Moon, K.; Im, H.-J.; Kwon, S. 3D graph contrastive learning for molecular property prediction. *Bioinformatics* **2023**, *39*, No. btad371.
- (16) Luo, S.; Chen, T.; Xu, Y.; Zheng, S.; Liu, T.-Y.; Wang, L.; He, D. One Transformer can Understand both 2D & 3D Molecular Data. 2022, arXiv:2210.01765. arXiv.org e-Printarchive. <https://arxiv.org/abs/2210.01765>.
- (17) Wang, L.; Liu, S.; Rong, Y.; Zhao, D.; Liu, Q.; Wu, S. MolSpectra: Pre-training 3D Molecular Representation with Multi-modal Energy Spectra. 2025, arXiv:2502.16284. arXiv.org e-Printarchive. <https://arxiv.org/abs/2502.16284>.
- (18) Huang, Z.; Fan, Z.; Shen, S.; Wu, M.; Deng, L. molecular representations for drug-related tasks through multi-view contrastive learning. *Bioinformatics* **2024**, *40*, ii190–ii197.
- (19) Tang, X.; Tran, A.; Tan, J.; Gerstein, M. B. MolLM: a unified language model for integrating biomedical text with 2D and 3D molecular representations. *Bioinformatics* **2024**, *40*, i357–i368.
- (20) Chithrananda, S.; Grand, G.; Ramsundar, B. ChemBERTa: Large-Scale Self-Supervised Pretraining for Molecular Property Prediction. 2020, arXiv:2010.09885. arXiv.org e-Printarchive. <https://arxiv.org/abs/2010.09885>.
- (21) Zang, X.; Zhao, X.; Tang, B. Hierarchical molecular graph self-supervised learning for property prediction. *Commun. Chem.* **2023**, *6*, No. 34.
- (22) Zhang, Z.; Liu, Q.; Wang, H.; Lu, C.; Lee, C.-K. In *Motif-Based Graph Self-Supervised Learning for Molecular Property Prediction*; Advances in Neural Information Processing Systems 34 NIPS, 2021; pp 15870–15882.
- (23) Hu, W.; Fey, M.; Ren, H.; Nakata, M.; Dong, Y.; Leskovec, J. Ogb-lsc: A Large-Scale Challenge for Machine Learning on Graphs. 2021, arXiv:2103.09430. arXiv.org e-Printarchive. <https://arxiv.org/abs/2103.09430>.
- (24) Wu, Z.; Ramsundar, B.; Feinberg, E. N.; Gomes, J.; Geniesse, C.; Pappu, A. S.; Leswing, K.; Pande, V. MoleculeNet: a benchmark for molecular machine learning. *Chem. Sci.* **2018**, *9*, 513–530.
- (25) Ramakrishnan, R.; Dral, P. O.; Rupp, M.; Von Lilienfeld, O. A. Quantum chemistry structures and properties of 134 kilo molecules. *Sci. Data* **2014**, *1*, No. 140022.
- (26) Fang, X.; Liu, L.; Lei, J.; He, D.; Zhang, S.; Zhou, J.; Wang, F.; Wu, H.; Wang, H. Geometry-enhanced molecular representation learning for property prediction. *Nat. Mach. Intell.* **2022**, *4*, 127–134.
- (27) Wang, J.; Guan, J.; Zhou, S. Molecular property prediction by contrastive learning with attention-guided positive sample selection. *Bioinformatics* **2023**, *39*, No. btad258.
- (28) Schütt, K. T.; Sauceda, H. E.; Kindermans, P.-J.; Tkatchenko, A.; Müller, K.-R. SchNet-a deep learning architecture for molecules and materials. *J. Chem. Phys.* **2018**, *148*, No. 241722, DOI: 10.1063/1.5019779.
- (29) Liu, Z.; Yang, D.; Wang, Y.; Lu, M.; Li, R. EGNN: Graph structure learning based on evolutionary computation helps more in graph neural networks. *Appl. Soft Comput.* **2023**, *135*, No. 110040.
- (30) Schütt, K.; Unke, O.; Gastegger, M. In *Equivariant Message Passing for the Prediction of Tensorial Properties and Molecular Spectra*; *Proceedings of the 38th International Conference on Machine Learning*, PMLR, 2021; pp 9377–9388.
- (31) Gasteiger, J.; Groß, J.; Günnemann, S. Directional Message Passing for Molecular Graphs. 2020, arXiv:2003.03123 arXiv.org e-Printarchive. <https://arxiv.org/abs/2003.03123>.
- (32) Coors, B.; Condurache, A. P.; Geiger, A. In *Spherenet: Learning Spherical Representations for Detection and Classification in Omnidirectional images*; Proceedings of the European conference on computer vision (ECCV) ECCV, 2018; pp 518–533.
- (33) Thölke, P.; De Fabritiis, G. Torchmd-net: Equivariant Transformers for Neural Network Based Molecular Potentials. 2022, arXiv:2202.02541 arXiv.org e-Printarchive. <https://arxiv.org/abs/2202.02541>.
- (34) Kreinovich, V.; Nguyen, H. T.; Ouncharoen, R. *How to Estimate Forecasting Quality: A System-Motivated Derivation of Symmetric Mean Absolute Percentage Error (SMAPE) and other Similar Characteristics*, 2014.
- (35) Landrum, G.; Tosco, P.; Kelley, B.; Cosgrove, D.; Vianello, R.; Kawashima, E.; Jones, G.; Dalke, A.; Cole, B.; Swain, M. *rdkit/rdkit: 03_1b1 (Q1 2023) Release2023*, Zenodo 2023.
- (36) Liao, Y.-L.; Wood, B.; Das, A.; Smidt, T. Equiformerv2: Improved Equivariant Transformer for Scaling to Higher-Degree Representations. 2023, arXiv:2306.12059. arXiv.org e-Printarchive. <https://arxiv.org/abs/2306.12059>.
- (37) Taheri, H.; Thrampoulidis, C. Fast convergence in learning two-layer neural networks with separable data. *Proc. AAAI Conf. Artif. Intell.* **2023**, *37*, 9944–9952.
- (38) Orio, M.; Pantazis, D. A.; Neese, F. Density functional theory. *Photosynth. Res.* **2009**, *102*, 443–453.
- (39) van den Oord, A.; Li, Y.; Vinyals, O. Representation Learning with Contrastive Predictive Coding. 2018, arXiv:1807.03748. arXiv.org e-Printarchive. <https://arxiv.org/abs/1807.03748>.
- (40) Vincent, P. A connection between score matching and denoising autoencoders. *Neural Comput.* **2011**, *23*, 1661–1674.



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